UKUDLA Machine Learning Certificate

Introduction to Machine Learning for Agri-Food Systems

Ayinde Mohammed Usman
University of the Western Cape

UKUDLA
African German Centre
for Sustainable and Resilient Food Systems and Applied Agricultural and Food Data Science

Supported by:
Federal Foreign Office

With funding from the:
Federal Ministry of Research, Technology and Space

With support from
Federal Ministry of Agriculture, Food and Regional Identity
by decision of the German Bundestag

DAAD Deutscher Akademischer Austauschdienst
German Academic Exchange Service

science, technology & innovation
Department:
Science, Technology and Innovation
REPUBLIC OF SOUTH AFRICA

NRF National Research Foundation

Table of contents

1. Introduction

2. Data Preparation

3. Feature Engineering

Session Roadmap

Part 1: Preparing the Data

Part 2: Feature Engineering

Session Roadmap

Part 1: Preparing the Data

1. The Crop Recommendation Dataset
2. Data Cleaning (Missing Values & Outliers)
3. Exploratory Data Analysis (EDA)

Part 2: Feature Engineering

Session Roadmap

Part 1: Preparing the Data

1. The Crop Recommendation Dataset
2. Data Cleaning (Missing Values & Outliers)
3. Exploratory Data Analysis (EDA)

Part 2: Feature Engineering

4. The 4-Step Framework
5. Creating Interaction & Ratio Features
6. Categorical Encoding & Collinearity

Goal of Feature Engineering

Transform a raw, messy agricultural dataset into a clean, mathematically enriched dataset ready for machine learning models.

UKUDLA

Learning Objectives

By the end of this session, you will be able to:

Data Preparation

- **Diagnose** missing values and justify when to impute vs. drop.
- **Detect and cap** extreme outliers using the IQR method.
- **Perform** EDA to discover biological interactions.

Feature Engineering

- **Execute** a rigorous 4-step framework.
- **Engineer** interaction (THI) and ratio (N:P) features.
- **Encode** categorical variables and filter out collinearity.

UKUDLA

Our Dataset: The Small Data Reality

Crop Recommendation Dataset

- **2,310 farm records**.
- **23 Features:** such as N, P, K, temperature, humidity, pH, rainfall, soil moisture.
- **Target:** Crop label (Classification).

UKUDLA

Our Dataset: The Small Data Reality

Crop Recommendation Dataset

- **2,310 farm records**.
- **23 Features:** such as N, P, K, temperature, humidity, pH, rainfall, soil moisture.
- **Target:** Crop label (Classification).

The Small Data Reality

We do not have "big data." If we feed raw features into a model, it may overfit.

Feature engineering is how we extract maximum signal from limited agricultural data.

UKUDLA

Handling Missing Values (Imputation)

The Problem: IoT sensors fail, or farmers forget to log data. We must fill the gaps.

Handling Missing Values (Imputation)

The Problem: IoT sensors fail, or farmers forget to log data. We must fill the gaps.

Why NOT the Mean?

If rainfall is $[0, 0, 0, 100]$, the mean is 25. But 25mm of rain never actually happened! The mean is heavily skewed by extreme weather.

Handling Missing Values (Imputation)

The Problem: IoT sensors fail, or farmers forget to log data. We must fill the gaps.

Why NOT the Mean?

If rainfall is $[0, 0, 0, 100]$, the mean is 25. But 25mm of rain never actually happened! The mean is heavily skewed by extreme weather.

Why the Median?

The median of $[0, 0, 0, 100]$ is 0. The median represents the "middle" reality of the field, making it robust and agronomically realistic.

UKUDLA

Handling Missing Values (Imputation)

The Problem: IoT sensors fail, or farmers forget to log data. We must fill the gaps.

Why NOT the Mean?

If rainfall is $[0, 0, 0, 100]$, the mean is 25. But 25mm of rain never actually happened! The mean is heavily skewed by extreme weather.

Why the Median?

The median of $[0, 0, 0, 100]$ is 0. The median represents the "middle" reality of the field, making it robust and agronomically realistic.

Jupyter Notebook: Median Imputation

```
df['rainfall'] = df['rainfall'].fillna(df['rainfall'].median())
```

UKUDLA

Handling Outliers (Winsorization)

The Problem: Storms or logging errors create extreme values that break model scaling. We do not delete them—we **cap** them.

Handling Outliers (Winsorization)

The Problem: Storms or logging errors create extreme values that break model scaling. We do not delete them—we **cap** them.

The Interquartile Range (IQR):

- $Q1 = 25\text{th}$ percentile
- $Q3 = 75\text{th}$ percentile
- $IQR = Q3 - Q1$
- **Upper Bound** $= Q3 + 1.5 \times IQR$

Any value above the upper bound is replaced by the upper bound.

UKUDLA

Handling Outliers (Winsorization)

The Problem: Storms or logging errors create extreme values that break model scaling. We do not delete them—we **cap** them.

The Interquartile Range (IQR):

- $Q1 = 25\text{th}$ percentile
- $Q3 = 75\text{th}$ percentile
- $IQR = Q3 - Q1$
- **Upper Bound** $= Q3 + 1.5 \times IQR$

Any value above the upper bound is replaced by the upper bound.

Jupyter Notebook: IQR Capping

```
Q1 = df['rainfall'].quantile(0.25)
Q3 = df['rainfall'].quantile(0.75)
IQR = Q3 - Q1
upper = Q3 + 1.5 * IQR
df['rainfall'] = np.where(df['rainfall'] >
upper,
upper, df['rainfall'])
```

Exploratory Data Analysis (EDA)

EDA is not about making pretty charts. It is about finding the missing biological links.

Exploratory Data Analysis (EDA)

EDA is not about making pretty charts. It is about finding the missing biological links.

- **Correlation Matrices:** If N, P, and K are highly correlated with each other, we have **multicollinearity**.

Exploratory Data Analysis (EDA)

EDA is not about making pretty charts. It is about finding the missing biological links.

- **Correlation Matrices:** If N, P, and K are highly correlated with each other, we have **multicollinearity**.
- **Scatter Plots:** Plot `temperature` vs `label`. Then plot `humidity` vs `label`.

Exploratory Data Analysis (EDA)

EDA is not about making pretty charts. It is about finding the missing biological links.

- **Correlation Matrices:** If N, P, and K are highly correlated with each other, we have **multicollinearity**.
- **Scatter Plots:** Plot `temperature` vs `label`. Then plot `humidity` vs `label`.
- **The Agronomic Insight:** Temperature alone might have a weak correlation with crop type. Humidity alone might be weak. But together? *High heat + High humidity = Deadly fungal diseases.*

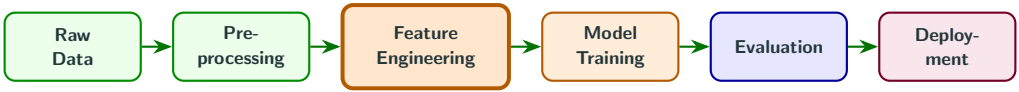
Exploratory Data Analysis (EDA)

EDA is not about making pretty charts. It is about finding the missing biological links.

- **Correlation Matrices:** If N, P, and K are highly correlated with each other, we have **multicollinearity**.
- **Scatter Plots:** Plot `temperature` vs `label`. Then plot `humidity` vs `label`.
- **The Agronomic Insight:** Temperature alone might have a weak correlation with crop type. Humidity alone might be weak. But together? *High heat + High humidity = Deadly fungal diseases.*

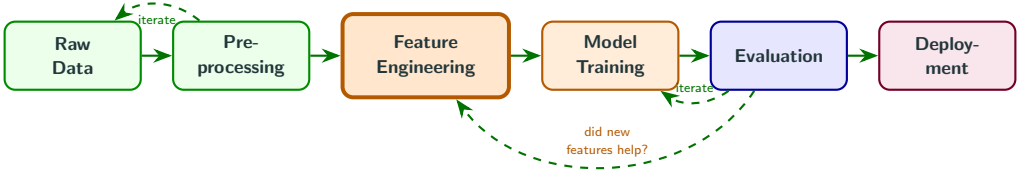
> **The Cue for Engineering:** When two variables interact to create an outcome that neither could produce alone, you must engineer a new feature to capture that interaction.

What is Feature Engineering?

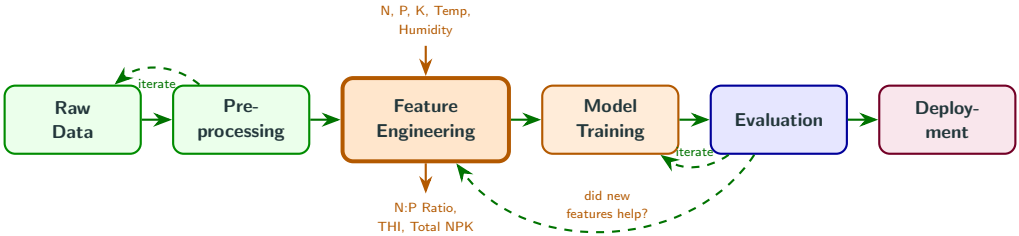


UKUDLA

What is Feature Engineering?

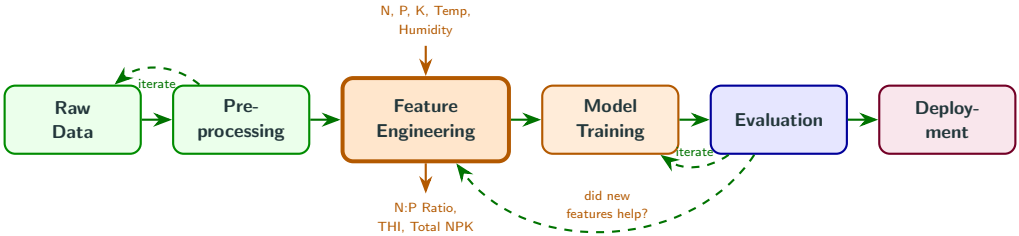


What is Feature Engineering?



UKUDLA

What is Feature Engineering?



The Definition

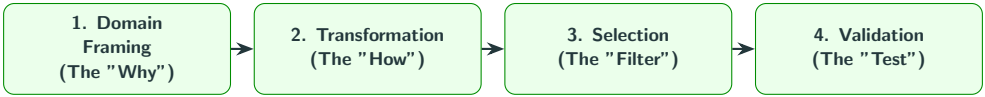
The process of using **domain knowledge** to create new variables from raw data to improve model performance.

The Core Philosophy

"Machines find the math. You find the meaning." Models cannot invent biological interactions on their own.

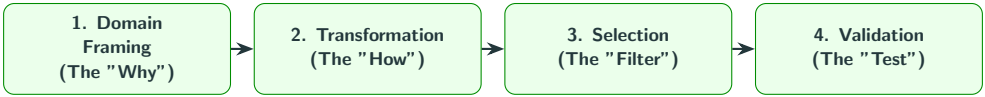
UKUDLA

How to Engineer Features: The 4-Step Framework



UKUDLA

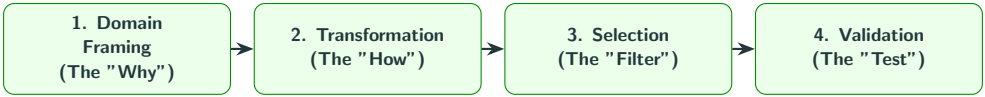
How to Engineer Features: The 4-Step Framework



- **Domain Framing:** What agronomic theory is missing?

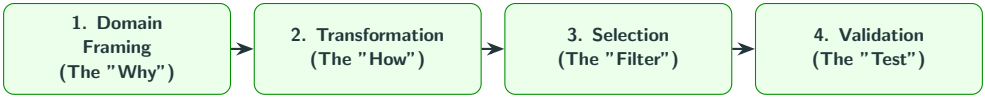
UKUDLA

How to Engineer Features: The 4-Step Framework



- **Domain Framing:** What agronomic theory is missing?
- **Transformation:** Write the mathematical formula.

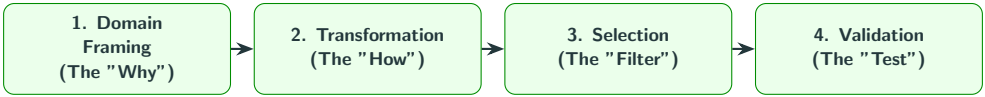
How to Engineer Features: The 4-Step Framework



- **Domain Framing:** What agronomic theory is missing?
- **Transformation:** Write the mathematical formula.
- **Selection:** Drop redundant features (Correlation > 0.85).

UKUDLA

How to Engineer Features: The 4-Step Framework



- **Domain Framing:** What agronomic theory is missing?
- **Transformation:** Write the mathematical formula.
- **Selection:** Drop redundant features (Correlation > 0.85).
- **Validation:** Did the model's error (RMSE/Accuracy) improve?

UKUDLA

Step 1: Domain Framing (The "Why")

The Question: What biological or agronomic theory explains the data?

Example: A raw temperature of 35°C is dangerous to maize **only if** humidity is high. High humidity prevents the plant from cooling itself via transpiration. Temperature alone is not enough.

Step 1: Domain Framing (The "Why")

The Question: What biological or agronomic theory explains the data?

Example: A raw temperature of 35°C is dangerous to maize **only if** humidity is high. High humidity prevents the plant from cooling itself via transpiration. Temperature alone is not enough.

Why this step matters

If you skip this and just multiply random columns, you risk overfitting. Domain framing ensures the new feature has a physical, causal basis in agriculture.

UKUDLA

Step 2: Transformation (The "How")

The Question: What is the mathematical formula?

Example: We transform Temperature (T) and Humidity (H) into the Temperature-Humidity Index (THI):

$$THI = (1.8T + 32) - (0.55 - 0.0055H)(1.8T - 26)$$

UKUDLA

Step 2: Transformation (The "How")

The Question: What is the mathematical formula?

Example: We transform Temperature (T) and Humidity (H) into the Temperature-Humidity Index (THI):

$$THI = (1.8T + 32) - (0.55 - 0.0055H)(1.8T - 26)$$

Why this step matters

This converts raw sensor readings (which are often on different scales) into a standardized, interpretable index. A THI > 79 means severe heat stress.

UKUDLA

Step 3: Selection (The "Filter")

The Question: Did this create redundancy (multicollinearity)?

Example: We now have `THI` and raw `temperature`. Do they contain the exact same information? We check the correlation matrix. If $|r| > 0.85$, we drop raw `temperature` and keep THI.

UKUDLA

Step 3: Selection (The "Filter")

The Question: Did this create redundancy (multicollinearity)?

Example: We now have `THI` and raw `temperature`. Do they contain the exact same information? We check the correlation matrix. If $|r| > 0.85$, we drop raw `temperature` and keep THI.

Why this step matters

Too many features increase complexity and confuse linear models. Filtering ensures we only keep the most informative variables.

UKUDLA

Step 4: Validation (The "Test")

The Question: Did it actually work?

Example: Train a Random Forest model with and without the new THI feature. Did the Root Mean Squared Error (RMSE) drop? Did Accuracy improve on the test set?

UKUDLA

Step 4: Validation (The "Test")

The Question: Did it actually work?

Example: Train a Random Forest model with and without the new THI feature. Did the Root Mean Squared Error (RMSE) drop? Did Accuracy improve on the test set?

Why this step matters

Feature engineering is a hypothesis. Validation is the experiment. If the model doesn't improve, the feature wasn't useful.

Method 1: Interaction Features

Combine two variables when their joint effect matters more than their individual effects.

The Formula: THI

$$THI = (1.8 \times \text{Temp}) + 32 - (0.55 - 0.0055 \times \text{Hum}) \times (1.8 \times \text{Temp} - 26)$$

UKUDLA

Method 1: Interaction Features

Combine two variables when their joint effect matters more than their individual effects.

The Formula: THI

$$THI = (1.8 \times \text{Temp}) + 32 - (0.55 - 0.0055 \times \text{Hum}) \times (1.8 \times \text{Temp} - 26)$$

Agronomic Rationale:

High temperature is stressful, but high humidity prevents transpiration. THI captures combined heat stress.

UKUDLA

Method 2: Ratio Features

Capture the balance between two variables. Absolute amounts can be misleading.

The Formula: N:P Ratio

$N_P_Ratio = \frac{N}{(P+1)}$ (Adding 1 prevents division by zero if P is 0).

UKUDLA

Method 2: Ratio Features

Capture the balance between two variables. Absolute amounts can be misleading.

The Formula: N:P Ratio

$N_P_Ratio = \frac{N}{(P+1)}$ (Adding 1 prevents division by zero if P is 0).

Agronomic Rationale:

A plant might have high absolute N, but if P is low, it cannot utilize the N. The ratio matters more than the raw numbers.

UKUDLA

Method 3: Aggregation Features

Combine multiple variables into a single summary metric.

The Formula: Total NPK

$$Total_NPK = N + P + K$$

Method 3: Aggregation Features

Combine multiple variables into a single summary metric.

The Formula: Total NPK

$$Total_NPK = N + P + K$$

Agronomic Rationale:

Instead of 3 separate nutrient values, this creates a "Total Fertility Index". It gives the model a quick snapshot of overall nutrient availability.

UKUDLA

Method 4: Domain Indices

Use established scientific formulas to translate raw sensors into biological metrics.

The Formula: Growing Degree Days (GDD)
$GDD = \max(\text{Temp} - T_{base}, 0)$ (Where T_{base} is the minimum temperature for growth, e.g., 10°C).

UKUDLA

Method 4: Domain Indices

Use established scientific formulas to translate raw sensors into biological metrics.

The Formula: Growing Degree Days (GDD)
$GDD = \max(\text{Temp} - T_{base}, 0)$ (Where T_{base} is the minimum temperature for growth, e.g., 10°C).

Agronomic Rationale:

GDD measures accumulated heat a crop receives. It predicts harvest timing much better than average daily temperature.

UKUDLA

Categorical Encoding

Models require numbers. They cannot calculate the distance between "Sandy" and "Clay".

Categorical Encoding

Models require numbers. They cannot calculate the distance between "Sandy" and "Clay".

One-Hot Encoding (Nominal)

Use when there is **no order** (e.g., Soil Type 1, 2, 3).

- Creates columns: `Soil_1`, `Soil_2`
- **Dummy Variable Trap:** Drop the first column. If it's not 1 and not 2, it *must* be 3.

Categorical Encoding

Models require numbers. They cannot calculate the distance between "Sandy" and "Clay".

One-Hot Encoding (Nominal)

Use when there is **no order** (e.g., Soil Type 1, 2, 3).

- Creates columns: `Soil_1`, `Soil_2`
- **Dummy Variable Trap:** Drop the first column. If it's not 1 and not 2, it *must* be 3.

Ordinal Encoding (Ordered)

Use when there is a **hierarchy**.

- Growth Stage: Seedling (1), Vegetative (2), Flowering (3).
- Preserves the mathematical order of the categories.

UKUDLA

The Danger of Too Many Features

Adding features increases complexity. If two features contain the same information, the model gets confused. This is **Multicollinearity**.

The Danger of Too Many Features

Adding features increases complexity. If two features contain the same information, the model gets confused. This is **Multicollinearity**.

Correlation Matrix
If the absolute correlation $|r| > 0.85$, drop one of the two features. They are redundant.

UKUDLA

The Danger of Too Many Features

Adding features increases complexity. If two features contain the same information, the model gets confused. This is **Multicollinearity**.

Correlation Matrix
If the absolute correlation $|r| > 0.85$, drop one of the two features. They are redundant.

VIF (Variance Inflation Factor)
A statistical measure. $VIF > 5$ indicates high collinearity. Drop the feature with the highest VIF.

UKUDLA

Open: `feature_engineering.ipynb`

What is done in the notebook:

Open: `feature_engineering.ipynb`

What is done in the notebook:

- Impute missing `rainfall` data using the median.
- Cap outliers in `rainfall` using the IQR method.
- Calculate the Temperature-Humidity Index (THI).
- Calculate the N:P Ratio and Total NPK.
- One-hot encode the `soil_type` column.
- Drop highly correlated features.

Key Takeaways

1. Domain knowledge drives feature engineering. Machines find math; you find meaning.

2. Always justify a new feature with agronomic theory (e.g., heat stress, nutrient balance).

3. More features $\neq$ better models. Check for collinearity ($|r| > 0.85$).

UKUDLA

Supported by

South African government and the National Research Foundation



science, technology & innovation

Department:
Science, Technology and Innovation
REPUBLIC OF SOUTH AFRICA



National Research Foundation

German government and the German Academic Exchange Service

Supported by:



Federal Foreign Office

With funding from the:



Federal Ministry of Research, Technology and Space

Supported by:



Federal Ministry of Agriculture, Food and Regional Identity

by decision of the German Bundestag

Project Manager:



Federal Office for Agriculture and Food

DAAD

Deutscher Akademischer Austauschdienst
German Academic Exchange Service

UKUDLA

References

- **Duboue, P.** (2020). *The Art of Feature Engineering*. Cambridge University Press.
- **Kuhn, M., & Johnson, K.** (2019). *Feature Engineering and Selection*. Chapman & Hall/CRC.
- **Zheng, A., & Casari, A.** (2018). *Feature Engineering for Machine Learning*. O'Reilly.
- **Kanter, J. M., & Veeramachaneni, K.** (2015). Deep Feature Synthesis. *IEEE DSAA*.